ANSWERS

1 **(a)** e.g. time (s), current (A), temperature (K), amount of substance (mol), luminous intensity (cdl)
1 each, max 3 …… B3 [3]

(b) density = mass / volume …… C1
unit of density: kg m^{-3} …… C1
unit of acceleration: m s^{-2} …… C1
unit of pressure: kg m^{-3} m s^{-2} m …… B1
kg m^{-1} s^{-2} …… B1 [5]
(*allow 4/5 for solution in terms of only dimensions*)

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2 10^{-9} …… B1
c …… B1
mega …… B1
tera …… B1 [4]

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3 **(a)** length, current, temperature, amount of substance, (luminous intensity)
any three, 1 each B3 [3]

(b) **(i)** *F*: $\mathrm{kg\,m\,s^{-2}}$ B1
ρ: $\mathrm{kg\,m^{-3}}$ B1
v: $\mathrm{m\,s^{-1}}$ B1 [3]

(ii) some working e.g. $\mathrm{kg\,m\,s^{-2}} = \mathrm{m^2\,kg\,m^{-3}}\,(\mathrm{m\,s^{-1}})^k$ M1
hence $k = 2$ A1 [2]

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4 **(a)** **(i)** scalar quantity has magnitude (allow size) B1
vector quantity has magnitude and direction B1 [2]

(ii) **1.** temperature: scalar B1 [1]
2. acceleration: vector B1 [1]
3. resistance: scalar B1 [1]

(b) *either* triangle / parallelogram with correct shape C1
tension = 14 .3 N (*allow ± 0.5 N*) A2 [3]

(*if > ±0.5 N but ≤ ±1 N, allow 1 mark*)
or $R = 25\cos 35°$ (C1)
$T = R\tan 35°$ (C1)
$T = 14.3$ N (A1)
or $T = 25\sin 35°$ (C2)
$T = 14.3$ N (A1)
or *R* and *T* resolved vertically and horizontally (C2)
leading to $T = 14.3$ N (A1)

5	**(a)**	scalar has only magnitude	B1	
		vector has magnitude and direction	B1	[2]
	(b)	kinetic energy, mass, power all three underlined	B1	[1]

6	**(a)**	**(i)**	V units: m^3 (*allow metres cubed or cubic metres*)	A1	[1]
		(ii)	Pressure units: $kg\,m\,s^{-2}$ / m^2 (*allow use of* $P = \rho gh$)	M1	
			Units: $kg\,m^{-1}\,s^{-2}$	A0	[1]
	(b)		V / t units: $m^3\,s^{-1}$	B1	
			Clear substitution of units for P, r^4 and l	M1	
			$C = \dfrac{\pi P r^4}{8 V t^{-1} l} = \dfrac{kg\,m^{-1}\,s^{-2}\,m^4}{m^3\,s^{-1}\,m}$		
			Units: $kg\,m^{-1}\,s^{-1}$	A1	[3]
			(*8 or* π *in final answer –1. Use of dimensions max 2/3*)		

7	**(e)**	**(i)**	arrow to the right of plane direction (about 4° to 24°)	B1	[1]
		(ii)	scale diagram drawn		
			or use of cosine formula $v^2 = 250^2 + 36^2 - 2 \times 250 \times 36 \times \cos 45°$		
			or resolving $v = [(36\cos 45°)^2 + (250 - 36\sin 45°)^2]^{1/2}$	C1	
			resultant velocity = 226 (220 – 240 for scale diagram) $m\,s^{-1}$		
			allow one mark for values 210 to 219 or 241 to 250 $m\,s^{-1}$		
			or use of formula ($v^2 = 51068$) v = 230 (226) $m\,s^{-1}$	A1	[2]

8	**(a)**		force: $kg\,m\,s^{-2}$	A1	[1]
	(b)	**(i)**	I^2: A^2 l: m x: m	C1	
			K: $kg\,m\,s^{-2}\,A^{-2}$	A1	[2]
		(ii)	curve of the correct shape (for inverse proportionality)	M1	
			clearly approaching each axis but never touching the axis	A1	[2]
		(iii)	curving upwards and through origin	A1	[1]

9	**(a)**		kelvin / K	B1	
			ampere / amp / A	B1	[2]
			[allow mole / mol and candela / Cd]		
	(b)	**(i)**	energy OR work = force × distance [allow any energy expression]	C1	
			units: $kg\,m\,s^{-2} \times m$ OR $kg\,(m\,s^{-1})^2$ for ½ mv^2 or mc^2 (ignore any numerical factor)	M1	
			$= kg\,m^2\,s^{-2}$	A0	[2]
		(ii)	units: ρ: $kg\,m^{-3}$ g: $m\,s^{-2}$ A: m^2 l_0: m	C1	
			C: $kg\,m^2\,s^{-2}$ / $kg^2\,m^{-6}\,m^2\,s^{-4}\,m^2\,m^3$ [any subject]	C1	
			$= kg^{-1}\,m\,s^2$ (allow $m\,s^2$ / kg)	A1	[3]

10 **(a)**	power = energy / time *or* work done / time	B1	
	force: $kg\,m\,s^{-2}$ (including from mg in mgh or Fv)		
	or kinetic energy ($\frac{1}{2}mv^2$): $kg\,(m\,s^{-1})^2$	B1	
	(distance: m and (time)$^{-1}$: s^{-1}) and hence power: $kg\,m\,s^{-2}\,m\,s^{-1} = kg\,m^2\,s^{-3}$	B1	[3]
(b)	Q/t : $kg\,m^2\,s^{-3}$	C1	
	A: m^2 and x: m and T: K	C1	
	correct substitution into $C = (Qx)/tAT$ or equivalent, or with cancellation	C1	
	units of C : $kg\,m\,s^{-3}\,K^{-1}$	A1	[4]
11 **(a)**	current, mass and temperature two correct 2/2, one omission or error 1/2	A2	[2]
(b)	σ: no units, V: m^3	C1	
	E_P: $kg\,m^2\,s^{-2}$	C1	
	C: $kg\,m^2\,s^{-2} \times m^{-3} = kg\,m^{-1}\,s^{-2}$	A1	[3]
12 **(a)**	temperature	B1	
	current	B1	[2]
	(allow amount of substance and luminous intensity)		
(b)	base units of force constant: $kg\,m\,s^{-2}\,m^{-1}$ or $kg\,s^{-2}$	B1	
	base units of time and mass: s and kg	C1	
	base units of C: $s\,(kg\,s^{-2}/kg)^{1/2}$ cancelling to show no units	B1	[3]
13 **(c)** **(i)**	arrow drawn up to the left of 7.5 N force approximately 5° to 40° to west of north	A1	[1]
(ii) **1.**	correct vector triangle or working to show magnitude of resultant force = 6.6 N allow 6.5 to 6.7 N if scale diagram	M1	[1]
2.	magnitude of acceleration = 6.6 / 0.75 [scale diagram: (6.5 to 6.7) / 0.75]	C1	
	= 8.8 $m\,s^{-2}$ [scale diagram: 8.7 – 8.9 $m\,s^{-2}$]	A1	[2]
(iii)	19° [use of scale diagram allow 17° to 21° (a diagram must be seen)]	B1	[1]

14	**(a)**	displacement / velocity / acceleration / momentum / etc. three correct (none wrong) 2, two correct (none or one wrong) 1	A2	[2]
	(b) (i)	Y = 70 N [allow 71 N as +½ small square on graph]	A1	[1]
	(ii)	$\theta = 90°$	M1	
		(for equilibrium) the direction of Y must be <u>opposite</u> to Z *or* using $Y \sin \theta = Z$, hence $\sin \theta = 70 / 70 = 1$, $\theta = 90°$	A1	[2]
	(iii) 1.	$Y \cos \theta = 160$ and $Y \sin \theta = 70$	C1	
		$\tan \theta = 70/160$ hence $\theta = 23.6°$ (24°)	A1	[2]
	2.	$Y = 160 / \cos 23.6°$ or $70 / \sin 23.6°$	C1	
		= 174.6 or 175 or 170 N	A1	[2]
		or:		
		$160^2 + 70^2 = Y^2$	(C1)	
		Y = 174.6 or 175 or 170 N	(A1)	
	(c)	(equilibrium not possible as) there is no vertical component from Y to balance Z	B1	[1]
15	**(a)**	ampere	B1	
		kelvin	B1	[2]
		(allow mole and candela)		
	(b) (i)	stress: N m^{-2}	C1	
		$kg\,m\,s^{-2}/m^2 = kg\,m^{-1}\,s^{-2}$	A1	[2]
	(ii)	Young modulus = stress / strain and <u>strain has no units</u> hence units: $kg\,m^{-1}\,s^{-2}$	B1	[1]
16	**(a)**	power = work / time or energy / time or (force × distance) / time	B1	
		$= kg\,m\,s^{-2} \times m\,s^{-1} = kg\,m^2\,s^{-3}$	A1	[2]
	(b)	power = VI [or V^2/R **and** $V = IR$ or I^2R **and** $V = IR$]	B1	
		(units of V:) $kg\,m^2\,s^{-3}\,A^{-1}$	B1	[2]
17	**(a)**	(work =) force × distance or force × displacement or (W =) $F \times d$	M1	
		units of work: $kg\,m\,s^{-2} \times m = kg\,m^2\,s^{-2}$	A1	[2]
	(b)	(p.d. =) $\dfrac{\text{work (done) or energy (transformed) (from electrical to other forms)}}{\text{charge}}$	B1	[1]
	(c)	$R = V/I$	B1	
		units of V: $kg\,m^2\,s^{-2}/A\,s$ **and** units of I: A	C1	
		or $R = P/I^2$ [or $P = VI$ **and** $V = IR$]	(B1)	
		units of P: $kg\,m^2\,s^{-3}$ **and** units of I: A	(C1)	
		or $R = V^2/P$	(B1)	
		units of V: $kg\,m^2\,s^{-2}/A\,s$ **and** units of P: $kg\,m^2\,s^{-3}$	(C1)	
		units of R: ($kg\,m^2\,s^{-2}/A^2\,s$ =) $kg\,m^2\,s^{-3}\,A^{-2}$	A1	[3]

18 **(a)** 150 or 1.5×10^2 Gm — A1 [1]

(b) distance = $2 \times (42.3 - 6.38) \times 10^6$ (= 7.184×10^7 m) — C1

(time =) $7.184 \times 10^7 / (3.0 \times 10^8) = 0.24$ (0.239) s — A1 [2]

(c) units of pressure P: $\text{kg m s}^{-2}/\text{m}^2 = \text{kg m}^{-1}\text{s}^{-2}$ — M1

units of density ρ: kg m^{-3} and speed v: m s^{-1} — M1

simplification for units of C: $C = v^2 \rho / P$ units: $(\text{m}^2\text{s}^{-2}\text{kg m}^{-3})/\text{kg m}^{-1}\text{s}^{-2}$
and cancelling to give no units for C — A1 [3]

(d) energy and power *(both underlined and no others)* — A1 [1]

(e) **(i)** vector triangle of correct orientation — M1

three arrows for the velocities in the correct directions — A1 [2]

(ii) length measured from scale diagram 5.2 ± 0.2 cm or components of boat speed determined parallel and perpendicular to river flow — C1

velocity = 2.6 m s^{-1} (allow ± 0.1 m s^{-1}) — A1 [2]

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19 **(a)** scalars: energy, power and time — A1

vectors: momentum and weight — A1 [2]

(b) **(i)** triangle with right angles between 120 m and 80 m, <u>arrows</u> in correct direction and result displacement from start to finish <u>arrow</u> in correct direction and labelled R — B1 [1]

(ii) **1.** average speed (= 200/27) = 7.4 m s^{-1} — A1 [1]

2. resultant displacement (= $[120^2 + 80^2]^{1/2}$) = 144 (m) — C1

average velocity (= 144/27) = 5.3(3) m s^{-1} — A1

direction (= $\tan^{-1}$ 80/120) = 34° (33.7) — A1 [3]

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20 **(a)** **(i)** force/area (normal to the force) — B1 [1]

(ii) ($p = F/A$ so) units: $\text{kg m s}^{-2}/\text{m}^2 = \text{kg m}^{-1}\text{s}^{-2}$ — A1 [1]

allow use of other correct equations:
e.g. ($\Delta p = \rho g \Delta h$ so) $\text{kg m}^{-3}\ \text{m s}^{-2}\ \text{m} = \text{kg m}^{-1}\text{s}^{-2}$
e.g. ($p = W/\Delta V$ so) $\text{kg m s}^{-2}\ \text{m}/\text{m}^3 = \text{kg m}^{-1}\text{s}^{-2}$

(b) units for m: kg, t: s and ρ: kg m^{-3} — C1

units of C: $\text{kg/s}\ (\text{kg m}^{-3}\ \text{kg m}^{-1}\ \text{s}^{-2})^{1/2}$ *or* units of C^2: $\text{kg}^2/\text{s}^2\ \text{kg m}^{-3}\ \text{kg m}^{-1}\ \text{s}^{-2}$ — C1

units of C: m^2 — A1 [3]

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21 **(a)**	kelvin, mole, ampere, candela *any two*	**B1**
(b)	use of resistivity = RA / l **and** $V = IR$ (to give $\rho = VA / Il$)	**C1**
	units of V: (work done / charge) kg m^2 s^{-2} (A s)$^{-1}$	**C1**
	units of resistivity: (kg m^2 s^{-3} A^{-1} A^{-1} m) = kg m^3 s^{-3} A^{-2}	**A1**
	or	
	use of $R = \rho L / A$ and $P = I^2R$ (gives $\rho = PA / I^2L$)	**(C1)**
	units of P: kg m^2 s^{-3}	**(C1)**
	units of resistivity: (kg m^2 s^{-3} × m^2) / (A^2 × m) = kg m^3 s^{-3} A^{-2}	**(A1)**

22	units of F: kg m s^{-2}	**C1**
	units of ρ: kg m^{-3} **and** units of v: m s^{-1}	**C1**
	units of K: kg m s^{-2} / [kg m^{-3} (m s^{-1})^2] = m^2	**A1**

23 **(a)** **(i)**	work (done) / time (taken) **or** energy (transferred) / time (taken)	**B1**
(ii)	Correct substitution of base units of all quantities into any correct equation for power.	**A1**
	Examples: ($P = E / t$ or W / t gives) kg m^2 s^{-2} / s = kg m^2 s^{-3} ($P = Fs / t$ or mgh / t gives) kg m s^{-2} m / s = kg m^2 s^{-3} ($P = ½mv^2 / t$ gives) kg (m s^{-1})^2 / s = kg m^2 s^{-3} ($P = Fv$ gives) kg m s^{-2} m s^{-1} = kg m^2 s^{-3} ($P = VI$ gives) kg m^2 s^{-2} A^{-1} s^{-1} A = kg m^2 s^{-3}	
(b) **(i)**	units of A: m^2 **and** units of T: K	**C1**
	units of k: kg m^2 s^{-3} / m^2 K^4 = kg s^{-3} K^{-4}	**A1**
(ii)	curve from the origin with increasing gradient	**B1**

24 (a)	a scalar has magnitude (only)	**B1**
	a vector has magnitude and direction	**B1**
(b)	power: scalar temperature: scalar momentum: vector	**B2**
	(two correct 1 mark, all three correct 2 marks)	
(c) (i)	arrow labelled R in a direction from 5° to 20° north of west	**B1**
(c)(ii)	$v^2 = 28^2 + 95^2 - (2 \times 28 \times 95 \times \cos 115°)$ **or** $v^2 = [(95 + 28\cos 65°)^2 + (28\sin 65°)^2]$	**C1**
	$v = 110\ \mathrm{ms^{-1}}$ ($109.8\ \mathrm{ms^{-1}}$)	**A1**
	or (scale diagram method)	
	triangle of velocities drawn	**(C1)**
	$v = 110\ \mathrm{m\,s^{-1}}$ (allow 108–112 $\mathrm{m\,s^{-1}}$)	**(A1)**

25 (a)	rate of change of momentum	**B1**
(b)	$\mathrm{kg\,m\,s^{-2}}$	**A1**
(c)	units for Q: A s **and** for r: m	**C1**
	units for $\varepsilon = (\mathrm{A\,s} \times \mathrm{A\,s})/(\mathrm{kg\,m\,s^{-2}} \times \mathrm{m^2})$ $= \mathrm{A^2\,kg^{-1}\,m^{-3}\,s^4}$	**A1**

26 (a) (i)	potential difference / current	**B1**
(ii)	$R = 4.0 \times 10^9\ (\Omega)$	**C1**
	$I = 0.60 / 4.0 \times 10^9 = 1.5 \times 10^{-10}$ (A)	**A1**
	$I = 150$ pA	
(b)	units of energy: $\mathrm{kg\,m^2\,s^{-2}}$	**C1**
	units of charge: A s	**C1**
	units of potential difference: ($\mathrm{kg\,m^2\,s^{-2}} / \mathrm{A\,s}$ =) $\mathrm{kg\,m^2\,A^{-1}\,s^{-3}}$	**A1**

27 (a)	scalar quantity has (only) magnitude	**B1**
	vector quantity has magnitude and direction	**B1**
(b)(i)	$E = F / Q$	**C1**
	$= \mathrm{kg\,m\,s^{-2}} / \mathrm{A\,s} = \mathrm{kg\,m\,A^{-1}\,s^{-3}}$	**A1**
(ii)	$b = Q / x^2E$	**C1**
	$= \mathrm{A\,s} / \mathrm{m^2\,kg\,m\,A^{-1}\,s^{-3}}$	
	$= \mathrm{A^2\,s^4\,kg^{-1}\,m^{-3}}$	**A1**

28 **(a)** **(i)** car uses 210 / 14 = 15 litres of fuel C1
volume reading = 45 litres .... A1 [2]

(ii) from 'full' to '3/4' mark .... B1 [1]

(b) **(i)** line/graph does not pass through ('empty, 0) / there is an intercept .... B1 [1]
(*do not allow 'non-linear'*)

(ii) (meter shows zero fuel when there is some left in the tank so)
acts as a 'reserve' .... B1 [1]

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29 **(a)** **(i)** *either* 1.55% *or* 1.6% …(not 1.5 or 2) .... A1 [1]

(ii) *either* 1.09% *or* 1.1% …(not 1.0 or 1) .... A1 [1]

(b) answer of {**(ii)** + 2 × **(i)**} to any number of sig. fig.
either 4.2% *or* 4.3% .... A1 [1]

(c) **(i)** *either* the value has more significant figures than the data
or uncertainty of ±0.4 renders more than 2 s.f. meaningless) .... B1 [1]

(ii) uncertainty in g = ±0.41 / ±0.42 to any number of s.f. C1
g = (9.8 ± 0.4) m s^{-2} .... A1 [2]

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30 **(a)** micrometer/screw gauge/digital callipers .... B1 [1]

(b) **(i)** look/check for zero error .... B1 [1]

(ii) take several readings M1
around the circumference/along the wire .... A1 [2]

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31 **(a)** **(i)** metre rule / tape (*not 'rule'*) B1 [1]

(ii) micrometer (screw gauge) / digital caliper B1 [1]

(iii) ammeter and voltmeter / ohmmeter / multimeter on 'ohm' setting B1 [1]

(b) **(i)** resistivity = RA / L C1
$= [7.5 \times \pi \times (0.38 \times 10^{-3})^2 / 4] / 1.75$ M1
$= 4.86 \times 10^{-7}\ \Omega$ m A0 [2]

(ii) (uncertainty in R =) [0.2 / 7.5] × 100 = 2.7%
<u>and</u> (uncertainty in L =) [3 / 1750] × 100 = 0.17% C1
(uncertainty in A =) 2 × (0.01 / 0.38) × 100 = 5.3 % C1
total = 8.13% C1

uncertainty = 0.395×10^{-7} (Ω m) A1 [4]
(*missing 2 factor in uncertainty in A, then allow max 3/4*)

(c) resistivity = $(4.9 \times 10^{-7} \pm 0.4 \times 10^{-7})\ \Omega$ m A1 [1]

32	**(a)**	$\dfrac{V}{t} = \dfrac{\pi P r^4}{8 C l}$		
		$C = [\pi \times 2.5 \times 10^3 \times (0.75 \times 10^{-3})^4] / (8 \times 1.2 \times 10^{-6} \times 0.25)$	C1	
		$= 1.04 \times 10^{-3}\ \mathrm{N\,s\,m^{-2}}$	A1	[2]
	(b)	$4 \times \%r$	C1	
		$\%C = \%P + 4 \times \%r + \%V/t + \%l$		
		$= 2\% + 5.3\% + 0.83\% + 0.4\%$ (= 8.6%)	A1	
		$\Delta C = \pm\, 0.089 \times 10^{-3}\ \mathrm{N\,s\,m^{-2}}$	A1	[3]
	(c)	$C = (1.04 \pm 0.09) \times 10^{-3}\ \mathrm{N\,s\,m^{-2}}$	A1	[1]
33	**(a)**	spacing = 380 or 3.8×10^2 pm	B1	[1]
	(b)	time = 24 × 3600 time = 0.086 (0.0864) Ms	B1	[1]
	(c)	time = distance / speed = $\dfrac{1.5 \times 10^{11}}{3 \times 10^{8}}$	C1	
		= 500 (s) = 8.3 min	A1	[2]
	(d)	momentum and weight	B1	[1]
34	**(a)**	$d = v \times t$	C1	
		$t = 0.2 \times 4$ (allow $t = 0.2 \times 2$)	C1	
		$d = 3 \times 10^8 \times 0.8 \times 10^{-6}$ OR $3 \times 10^8 \times 0.4 \times 10^{-6}$	C1	
		d = 240 m hence distance from source to reflector = 120 m	A1	[4]
	(b)	speed of sound 300 cf speed of light 3×10^8 OR time = 240 / 300 (= 0.8) OR time = 120 / 300 (= 0.4)	C1	
		sound slower by factor of 10^6 OR time for one division 0.8 / 4 OR time for one division 0.4 / 2	C1	
		time base setting 0.2 s cm^{-1} [unit required]	A1	[3]
35		volume = $\pi\,(14 \times 10^{-3})^2 \times 12 \times 10^{-3}$ ($=7.389 \times 10^{-6}$ m^3) density = mass / volume [any subject] mass = $6.8 \times 10^3 \times 7.389 \times 10^{-6} = 0.0502$	C1	
		weight = mg $= 0.0502 \times 9.81 = 0.49$ N (mark not awarded if not to **two** s.f.)	C1 A1	[4]

36	**(a)**	SI units for T: s, R: m and M: kg (or seen clearly in formula)	C1	
		$K = T^2M / R^3$ units: $\text{s}^2\ \text{kg m}^{-3}$ (allow $\text{s}^2\ \text{kg / m}^3$ or $\frac{\text{s}^2\text{kg}}{\text{m}^3}$)	A1	[2]
	(b)	% uncertainty in K: 1% (for T) + 3% (for R) + 2% (for M) OR = 6%	C1	
		$K = [(86400)^2 \times 6 \times 10^{24}] / (4.23 \times 10^7)^3 = 5.918 \times 10^{11}$	C1	
		6% of $K = 0.355 \times 10^{11}$	C1	
		$K = (5.9 \pm 0.4) \times 10^{11}$ (SI units) correct power of ten required for both	A1	[4]
		[incorrect % value then max. 1]		
37	**(a)**	$\rho = m / V$	C1	
		$V = (\pi d^2/4) \times t = 7.67 \times 10^{-7}\ \text{m}^3$		
		$\rho = (9.6 \times 10^{-3})/[\pi(22.1/2 \times 10^{-3})^2 \times 2.00 \times 10^{-3}]$	C1	
		$\rho = 12513\ \text{kg m}^{-3}$ (allow 2 or more s.f.)	A1	[3]
	(b) (i)	$\Delta\rho/\rho = \Delta m/m + \Delta t/t + 2\Delta d/d$	C1	
		= 5.21% + 0.50% + 0.905% [or correct fractional uncertainties]	C1	
		= 6.6% (6.61%)	A1	[3]
	(ii)	$\rho = 12\,500 \pm 800\ \text{kg m}^{-3}$	A1	[1]
38	**(a)**	pressure = force / area (normal to the force) [clear ratio essential]	B1	[1]
	(b) (i)	$P = mg / A = (5.09 \times 9.81) / A$	C1	
		$A = (\pi d^2/4) = \pi \times (9.4 \times 10^{-2})^2 / 4$ (= $0.00694\ \text{m}^2$)	C1	
		$P = 49.93 / 0.00694$		
		= 7200 (7195) Pa (minimum of 2 s.f. required)	A1	[3]
	(ii)	$\Delta P / P = \Delta m / m + 2\Delta d / d$	C1	
		$= 0.01 / 5.09 + (2 \times 0.1) / 9.4$ (= 0.0020 + 0.021 or 2.3%)	C1	
		ΔP = 170 (165 to 167) Pa	A1	[3]
	(iii)	$P = 7200 \pm 200$ Pa	A1	[1]
39	**(a)**	<u>random</u> error (in the measurements) of the length OR resistance	B1	[1]
	(b)	gradient = (3.6 – 1.9) / (0.8 – 0.4)	C1	
		= 4.25	A1	[2]
	(c)	$R = \rho l / A$	C1	
		ρ = gradient $\times$ area $= 4.25 \times 0.12 \times 10^{-6}$	C1	
		$= 5.1(0) \times 10^{-7}\ \Omega\,\text{m}$	A1	[3]

40	**(a)**	**(i)**	amplitude scale reading 2.2 (cm)	C1	
			amplitude = 2.2×2.5 = 5.5 mV	A1	[2]
		(ii)	time period scale reading = 3.8 (cm)	C1	
			time period = $3.8 \times 0.5 \times 10^{-3}$ = 0.0019 (s)	C1	
			frequency f = 1 / 0.0019 = 530 (526) Hz	A1	[3]
		(iii)	uncertainty in reading = ± 0.2 in 3.8 (cm) or 5.3% or 0.2 in 7.6 (cm) or 2.6% [allow other variations of the distance on the *x*-axis]	M1	
			actual uncertainty = 5.3% of 526 = 27.7 or 28 Hz *or* 2.6% of 526 = 13 or 14	A1	[2]
	(b)		frequency = 530 ± 30 Hz or 530 ± 10 Hz	A1	[1]

41	**(a)**	**(i)**	diameter and extension: micrometer (screw gauge) or digital calipers	B1	
			length: tape measure or metre rule	B1	
			load: spring balance or Newton meter	B1	[3]
		(ii)	to reduce the effect of random errors **or** to plot a graph to check for zero error in measurement of extension **or** to see if limit of proportionality is exceeded	B1	[1]

42	**(a)**		metre rule / tape measure	B1
	(b)	**(i)**	$v = [(1.8 \times 126 \times 10^{-2}) / 5.1 \times 10^{-3}]^{1/2}$	C1
			$= 21.1\ (\mathrm{m\,s^{-1}})$	A1
		(ii)	percentage uncertainty = 4% **or** fractional uncertainty = 0.04	C1
			$\Delta v = 0.04 \times 21.1$	
			$= 0.84$	C1
			$v = 21.1 \pm 0.8\ (\mathrm{m\,s^{-1}})$	A1

43	**(a)**	**(i)**	$(50 \text{ to } 200) \times 10^{-3}$ kg or (0.05 to 0.2) kg	B1	[1]
		(ii)	(50 to 300) $\mathrm{cm^3}$	B1	[1]
	(b)		density = mass / volume or $\rho = M / V$	C1	
			$V = [\pi(0.38 \times 10^{-3})^2 \times 25.0 \times 10^{-2}]/4$ (= 2.835×10^{-8} $\mathrm{m^3}$)	C1	
			$\rho = (0.225 \times 10^{-3}) / 2.835 \times 10^{-8}$ $= 7940\ (\mathrm{kg\,m^{-3}})$	A1	
			$\Delta\rho/\rho = 2(0.01/0.38) + (0.1/25.0) + (0.001/0.225)$ [= 0.061] *or* $\%\rho$ = 5.3% + 0.40% + 0.44% (= 6.1%)	C1	
			$\Delta\rho = 0.061 \times 7940 = 480\ (\mathrm{kg\,m^{-3}})$		
			density = $(7.9 \pm 0.5) \times 10^3\ \mathrm{kg\,m^{-3}}$ or $(7900 \pm 500)\ \mathrm{kg\,m^{-3}}$	A1	[5]

44 **(a)** systematic: the reading is larger or smaller than (or varying from) the true reading by a constant amount	B1	
random: scatter in readings about the true reading	B1	[2]
(b) precision: the size of the smallest division (on the measuring instrument) *or* 0.01 mm for the micrometer	B1	
accuracy: how close (diameter) value is to the true (diameter) value	B1	[2]

45 **(a)** (density =) mass / volume		
(b) **(i)** $d = [(6 \times 7.5)/(\pi \times 8100)]^{1/3}$		
$= 0.12(1)$ m	A1	[1]
(ii) percentage uncertainty = (4 + 5) / 3 (= 3%) *or* fractional uncertainty = (0.04 + 0.05) / 3 (= 0.03)	C1	
absolute uncertainty (= 0.03 × 0.121) = 0.0036	C1	
$d = 0.121 \pm 0.004$ m	A1	[3]

46 **(a)** (stress =) force / area **or** kg m s^{-2} / m^2	**B1**
= kg m^{-1} s^{-2}	**A1**
(b) **(i)** $0.58 = 2\pi \times [(4 \times 0.500 \times 0.600^3)/(E \times 0.0300 \times 0.00500^3)]^{0.5}$	**C1**
$E = [4\pi^2 \times 4 \times 0.500 \times (0.600)^3] / [(0.58)^2 \times 0.0300 \times (0.00500)^3]$	**C1**
$= 1.35 \times 10^{10}$ (Pa)	
= 14 (13.5) GPa	**A1**
(ii) **1** (accuracy determined by) the closeness of the value(s)/measurement(s) to the true value	**B1**
(precision determined by) the range of the values/measurements	**B1**
2 l is (cubed so) 3 × (percentage/fractional) uncertainty **and** T is (squared so) 2 × (percentage / fractional) uncertainty **and** (so) l contributes more	**B1**

47 **1.** precision is determined by the range in the measurements/values/readings/data/results	**B1**
2. metre rule measures to ± 1 mm and micrometer to ± 0.01 mm (so there is less (percentage) uncertainty/random error)	**B1**

48 (a) (i) micrometer (screw gauge)/digital calipers **B1**

(a) (ii) take several readings (and average) **M1**

along the wire **or** around the circumference **A1**

(b) (i) $\sigma = 4 \times 25 / [\pi \times (0.40 \times 10^{-3})^2] = 1.99 \times 10^8$ N m^{-2} **A1**

or $\sigma = 25 / [\pi \times (0.20 \times 10^{-3})^2] = 1.99 \times 10^8$ N m^{-2}

(b) (ii) $\%F = 2\%$ **and** $\%d = 5\%$ **C1**

or $\Delta F / F = \dfrac{0.5}{25}$ **and** $\Delta d / d = \dfrac{0.02}{0.4}$

$\%\sigma = 2\% + (2 \times 5\%)$ **A1**

or $\%\sigma = [0.02 + (2 \times 0.05)] \times 100$

$\%\sigma = 12\%$

(b) (iii) absolute uncertainty $= (12 / 100) \times 1.99 \times 10^8$ **C1**

$= 2.4 \times 10^7$

$\sigma = 2.0 \times 10^8 \pm 0.2 \times 10^8$ N m^{-2} **or** $2.0 \pm 0.2 \times 10^8$ N m^{-2} **A1**

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49 **(a)** **(i)** zero error **or** wrongly calibrated scale **B1**

(ii) reading scale from different angles **or** wrongly interpolating between scale readings/divisions **B1**

(b) **(i)** $P = V^2/R$ **or** $P = VI$ and $V = IR$ **C1**

$P = 5.0^2 / 125$ **or** 5.0×0.04 **or** $(0.04)^2 \times 125$ **A1**

$= 0.20$ W

(ii) $\%V = 2\%$ **or** $\Delta V / V = 0.02$ **C1**

$\%P = (2 \times 2\%) + 3\%$ **or** $\%P = (2 \times 0.02 + 0.03) \times 100$ **A1**

$= 7\%$

(iii) absolute uncertainty in $P = (7 / 100) \times 0.20$ **C1**

$= 0.014$

power $= 0.20 \pm 0.01$ W **or** $(2.0 \pm 0.1) \times 10^{-1}$ W **A1**

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50 **(a)** (velocity =) change in displacement / time (taken) **B1**

(b)(i) $k = [1.29 \times (3.3 \times 10^2)^2] / 9.9 \times 10^4$ **C1**

$= 1.4$ **A1**

(ii) percentage uncertainty $= (3 \times 2) + 4 + 2$ (= 12%) **C1**

or

fractional uncertainty $= (0.03 \times 2) + 0.04 + 0.02$ (= 0.12)

$\Delta k = 0.12 \times 1.42$ **C1**

$= 0.17$ (*allow to 1 significant figure*)

$k = 1.4 \pm 0.2$ **A1**

51	**(a)**	absolute uncertainty $= (1.6 / 100) \times 0.0125$ $= 2 \times 10^{-4}$ m	**A1**
	(b)(i)	$p = (4 \times 0.38) / (\pi \times 0.0125^2)$ $= 3100$ N m^{-2}	**A1**
	(ii)	percentage uncertainty $= 2.8 + (2 \times 1.6)$ (= 6%) **or** fractional uncertainty $= 0.028 + (2 \times 0.016)$ (= 0.06)	**C1**
		absolute uncertainty $= 0.06 \times 3100$ $= 190$ N m^{-2} (*allow to 1 significant figure*)	**A1**

52	**(a)(i)**	mass in range 1–20 g	**A1**
	(ii)	wavelength in range 1×10^{-8} m to 4×10^{-7} m	**A1**
	(b)(i)	$T = 2\pi \times (200 \times 10^{-3} / 25)^{0.5}$ $= 0.56$ s	**A1**
	(ii)	percentage uncertainty = (2% + 8%) / 2 (= 5%) **or** fractional uncertainty = (0.02+0.08) / 2 (= 0.05)	**C1**
		$\Delta T = 0.56 \times 0.05$ $= 0.028$ (s)	**C1**
		$T = (0.56 \pm 0.03)$ s	**A1**

53	**(a)**	(work =) force $\times$ displacement	**C1**
		units: kg m s^{-2} $\times$ m = kg m^2 s^{-2}	**A1**
	(b)(i)	units of Q: As	**C1**
		units of C: kg^{-1} m^{-2} A^2 s^4	**A1**
	(ii)	**1.** e.g. reading scale from different angles (wrongly) interpolating between scale readings/divisions	**B1**
		2. e.g. zero error wrongly calibrated scale	**B1**

54.

(a)(i)	wavelength = 8.5×10^{-5} m
(a)(ii)	$f = v / \lambda$ **or** c / λ
	$= 3.0 \times 10^{8} / 8.5 \times 10^{-5}$ (= 3.5×10^{12}) = 3.5 THz
(a)(iii)	infrared
(b)	(implied) percentage uncertainty in I = 4% **or** (implied) fractional uncertainty in I = 0.04
	percentage uncertainty in E = 5% + (4% $\times$ 2) = 13%

55.

(a)	mass / volume
(b)(i)	absolute uncertainty = $4.0 \times (5 / 100)$ = ($\pm$) 0.2 cm
(b)(ii)	percentage uncertainty = 2 + 4 + (5 $\times$ 2)
	= ($\pm$) 16%
(c)	$p = F / A$ **or** $p = W / A$
	$p = (19.5 \times 10^{-3} \times 9.81) / (4.0 \times 10^{-2})^2$
	= 120 Pa

56.

(a)(i)	two correct scalar quantities e.g. time, mass, distance, temperature
	two correct vector quantities e.g. force, acceleration, velocity, displacement
(a)(ii)	magnitude
	unit
(b)(i)	north component of velocity = 11 m s^{-1}
	east component of velocity = 7.5 m s^{-1}
(b)(ii)	velocity = 7.5 – 2.7 = 4.8 m s^{-1}
(b)(iii)	velocity = $\sqrt{(11^2 + 4.8^2)}$
	= 12 m s^{-1}
(b)(iv)	angle = $\tan^{-1}$ (4.8 / 11)
	= 24°

57.

(a)	mass / volume
(b)(i)	(vernier/digital) calipers
(b)(ii)	percentage uncertainty = (0.0004 / 0.0420) × 100 = 1%
(c)(i)	kg m^{-3} = kg × m^n / m **or** kg m^{-3} = kg × m^n × m^{-1}
	$-3 = n - 1$ and (so) $n = -2$
(c)(ii)	$(\Delta\rho / \rho) = (\Delta M / M) + 2(\Delta r / r) + (\Delta L / L)$
	percentage uncertainty = [(0.001 / 1.072) + 2 × (0.0004 / 0.0420) + (0.0001 / 0.1242)] (× 100)
	= 0.09% + 2 × 0.95% + 0.08% = 2%
(c)(iii)	ρ = (1.072 × 0.0420^{-2}) / (2.094 × 0.1242) = 2337 (kg m^{-3})
	$\Delta\rho$ = 0.021 × 2337 = 49 (kg m^{-3})
	ρ = (2340 ± 50) kg m^{-3}

58.

(a)	10^{12}
	pico (p)
(b)	ampere and metre both underlined (and no other units underlined)
(c)(i)	percentage uncertainty $= 3.5 + (3.0 \times 2) + 2.5 + 2.0$
	$= 14\%$
(c)(ii)	absolute uncertainty $= 4.1 \times 10^{-7} \times 14 / 100$
	$= 6 \times 10^{-8}\ \Omega\ \text{m}$

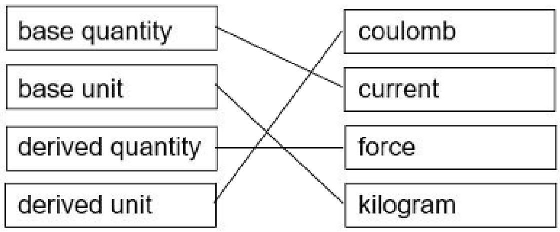
59.

(a)	only ampere and kilogram underlined
(b)(i)	percentage uncertainty $= 2.1 + 0.6 + (1.5 \times 2)$
	$= 5.7\%$
(b)(ii)	absolute uncertainty $= (5.7 / 100) \times 8.245 \times 10^9$
	$(= 4.7 \times 10^8$ Pa or 0.47×10^9 Pa)
	$E = (8.2 \pm 0.5) \times 10^9$ Pa

60.

(a)	$\rho = m / V$
	$V = (4/3) \times \pi \times r^3$ $= (4/3) \times \pi \times (3.42/2)^3$ $(= 20.9\ \text{cm}^3)$
	$\rho = 67 / 20.9$ $= 3.2\ \text{g cm}^{-3}$
(b)	$\%\rho = \%m + 3 \times \%d$ $= [(2/67) \times 100] + [3 \times (0.02/3.42) \times 100]$
	$= 3.0\% + 3 \times 0.58\%$ $= 4.7\%$ or 5%

61.

(a)	 any two joined correctly
	all four joined correctly
(b)(i)	the measurements have a small range
(b)(ii)	(average of the) measurements not close to the true value
(c)(i)	percentage uncertainty = (3 + 5 + 4) / 2
	= 6%
(c)(ii)	absolute uncertainty = (6 / 100) × 15.0 = 0.9 m s^{-1}

62.

(a)	force × displacement in the direction of the force
(b)	$P = Fs / t$ = (kg m s^{-2} × m) / s
	= kg m^2 s^{-3}
(c)(i)	$84 \times 10^3 = v^3 \times 0.56$
	v = 53 m s^{-1}
(c)(ii)	percentage uncertainty = (5% + 7%) / 3 (= 4%) **or** fractional uncertainty = (0.05 + 0.07) / 3 (= 0.04)
	absolute uncertainty = 0.04 × 53 = (±) 2 m s^{-1}

63.

(a)	(SI base unit of α =) $m^3\ s^{-1} \times m / m^4 = s^{-1}$
(b)	(percentage uncertainty =) $3 + 4 + 2 \times 4 = 15$ (%)
(c)	$\alpha = QL / r^4$ $= 2.72 \times 10^{-8} \times 2.5 \times 10^{-2} / (7.1 \times 10^{-5})^4$ $= 2.7 \times 10^7$
	absolute uncertainty $= 0.15 \times [2.7 \times 10^7]$ $= 0.4 \times 10^7$
	$\alpha = (2.7 \pm 0.4) \times 10^7\ s^{-1}$

64.

(a)	only ampere and kelvin underlined
(b)	initial speed / velocity is zero
	(non-zero magnitude of) acceleration is constant / uniform (and in a straight line)
(c)(i)	$a = 2.75^2 / (2 \times 3.89)$ $= 0.97\ m\ s^{-2}$
(c)(ii)	percentage uncertainty $= (2 \times 0.8) + 0.5$
	$= 2.1\%$
(c)(iii)	absolute uncertainty $= (2.1 / 100) \times 0.97$ $= 0.02\ m\ s^{-2}$

65.

(a)(i)	work done per unit time
(a)(ii)	($P = W / t$ gives) units: $kg\ m^2\ s^{-2} / s = kg\ m^2\ s^{-3}$
(b)	($I = P / A$ so) units of I: $kg\ m^2\ s^{-3} / m^2$ or $kg\ s^{-3}$
	units of f: s^{-1} **and** units of A: m **and** units of v: $m\ s^{-1}$
	units of k: $kg\ s^{-3} / [(s^{-1})^2\ m^2\ m\ s^{-1}]$ $= kg\ m^{-3}$

66.

(a)(i)	force / area (normal to the force)
(a)(ii)	($p = F / A$ so units are) kg m s^{-2} / m^2 = kg m^{-1} s^{-2}
(b)	unit of R: m **and** unit of t: s **and** unit of L: m
	unit of ρ: kg m^{-3} **or** $\rho = m / V$
	base units of k: (kg m^{-1} s^{-2} × m^4 × kg m^{-3} × s) / (kg × m) = kg m^{-1} s^{-1}
(c)	R contributes 4 × 2% **or** 8% (and L contributes 2%) so R contributes more (to the percentage uncertainty in k)

67.

(a)	$t = \sqrt{(2s / g)}$ $= \sqrt{[(2 \times 36) / 9.81]}$
	= 2.7 s
(b)	• reaction time between hearing the splash and stopping the stop-watch • the sound (of the splash) takes time to reach the student **or** the stone hits the water at a different time to the sound being heard **or** the sound (of the splash) has to travel to the student • the student might not let go of the stone from ground level • the student might not let go of the stone and start the stop-watch at the same time • stop-watch may not be properly calibrated / has a zero error • (local value of) g is not (exactly) 9.81 (m s^{-2}) • stone given initial velocity / initial velocity not zero • stone does not fall (exactly) vertically / in a straight line *Any three points, 1 mark each*
(c)	precise: results are close together / have little scatter
	not accurate: the values are not close to / 50% different / (very) different from the true value

68.

(a)	scalar and vector have magnitude
	vector has direction (and scalar does not have direction)
(b)(i)	$r = [(3 \times 28) / 4\pi]^{1/3}$ = 1.9 cm
(b)(ii)	percentage uncertainty in V = (0.5 / 28) × 100 (= 1.79%)
	percentage uncertainty in r = 1.79 / 3 = 0.6%

69.

(a)	charge **and** power only ticked
(b)(i)	$\%D = 0.4\%$ **and** $\%L = 0.6\%$
(b)(ii)	$\rho = (4 \times 0.247) / [\pi \times (26.2 \times 10^{-3})^2 \times 0.162]$
	$\rho = 2.83 \times 10^3$ kg m^{-3}
(b)(iii)	percentage uncertainty $= 0.4 + (2 \times 0.4) + 0.6$
	$= 1.8\%$